

Sampling process

S18

①. Find the Nyquist rate and Nyquist interval for following signals.

i). $m(t) = \frac{1}{2\pi} \cos(4000\pi t) \cdot \cos(1000\pi t)$.

ii). $m(t) = \frac{\sin 500\pi t}{\pi t}$.

i). $m(t) = \frac{1}{2\pi} \cdot \cos(4000\pi t) \cdot \cos(1000\pi t)$.

$$= \frac{1}{2\pi} \times \frac{2}{2} (\cos(4000\pi t + 1000\pi t) + \cos(4000\pi t - 1000\pi t))$$

$$= \frac{1}{4\pi} \{ \cos 3000\pi t + \cos 5000\pi t \}$$

$$= \frac{1}{4\pi} \{ \cos 2\pi f_1 t + \cos 2\pi f_2 t \}$$

$$f_1 = 1500, \quad f_2 = 2500 \text{ Hz}$$

$$\text{Highest frequency Component} = f_2 = 2500 \text{ Hz}$$

$$\begin{aligned} \text{NR} = \text{Nyquist rate} &= 2 \times 2500 \\ &= 5000 \text{ Hz} \end{aligned}$$

$$\text{Nyquist interval} = \frac{1}{\text{NR}} = \frac{1}{5000} = 0.2 \text{ msec}$$

ii). $m(t) = \frac{\sin(500\pi t)}{\pi t} = \frac{\sin(2\pi f_1 t)}{\pi t}$

$$f_1 = \frac{500}{2} = 250 \text{ Hz}$$

$$\text{NR} = \text{Nyquist rate} = 2 \times 250 = 500 \text{ Hz}$$

$$\text{Nyquist interval} = \frac{1}{\text{NR}} = \frac{1}{500} = 2 \text{ msec}$$

②. A waveform consisting of a sinusoidal of fundamental frequency 500Hz and its third harmonic component has an amplitude of 0.1 times the fundamental component is sampled at the rate of $10,000$ samples/sec ideally. Sketch a neat figure of the spectrum of sampled waveform showing first four components.

$$x(t) = \sin 2\pi f_c t. \quad \text{--- (1)} \quad , f_s = 10,000 \text{ samples/sec}$$

by taking fourier transform of eq-1.

$$x(f) = \frac{1}{2j} [\delta(f - f_c) - \delta(f + f_c)]$$

$$= \frac{1}{2j} [\delta(f - f_c) - \delta(f + f_c)]$$

$$f_1 = 500$$

$$\text{Amp} = 1V$$

$$f_3 = 1500$$

$$\text{Amp} = 0.1V$$

$$x(t) = \sin(2\pi f_1 t) + 0.1 \sin(2\pi f_3 t)$$

$$= \sin(2\pi \times 500 t) + 0.1 \sin(2\pi \times 1500 t)$$

$$x(f) = \frac{1}{2j} [\delta(f - 500) - \delta(f + 500)] + 0.1 \{ \delta(f - 1500) - \delta(f + 1500) \}$$

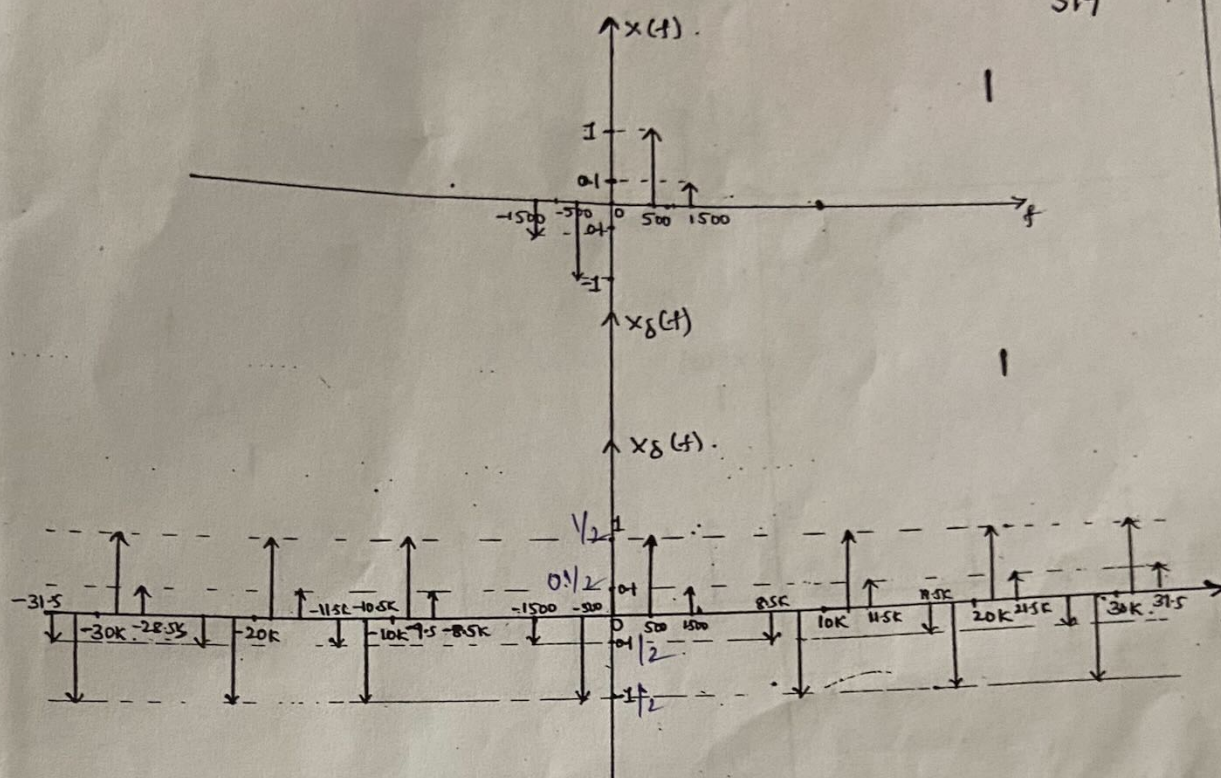
$$x(f) = \frac{1}{2j} \{ [\delta(f - 500) - \delta(f + 500)] + 0.1 [\delta(f - 1500) - \delta(f + 1500)] \}$$

Sampled signal.

$$x_s(f) = f_s \sum_{n=-\infty}^{\infty} x(f - n f_s)$$

$$= f_s x(f) + f_s x(f \pm f_s) + f_s x(f \pm 2f_s) + f_s x(f \pm 3f_s)$$

$$= 10,000 [x(f) + x(f \pm 10\text{kHz}) + x(f \pm 20\text{kHz}) + x(f \pm 30\text{kHz})]$$



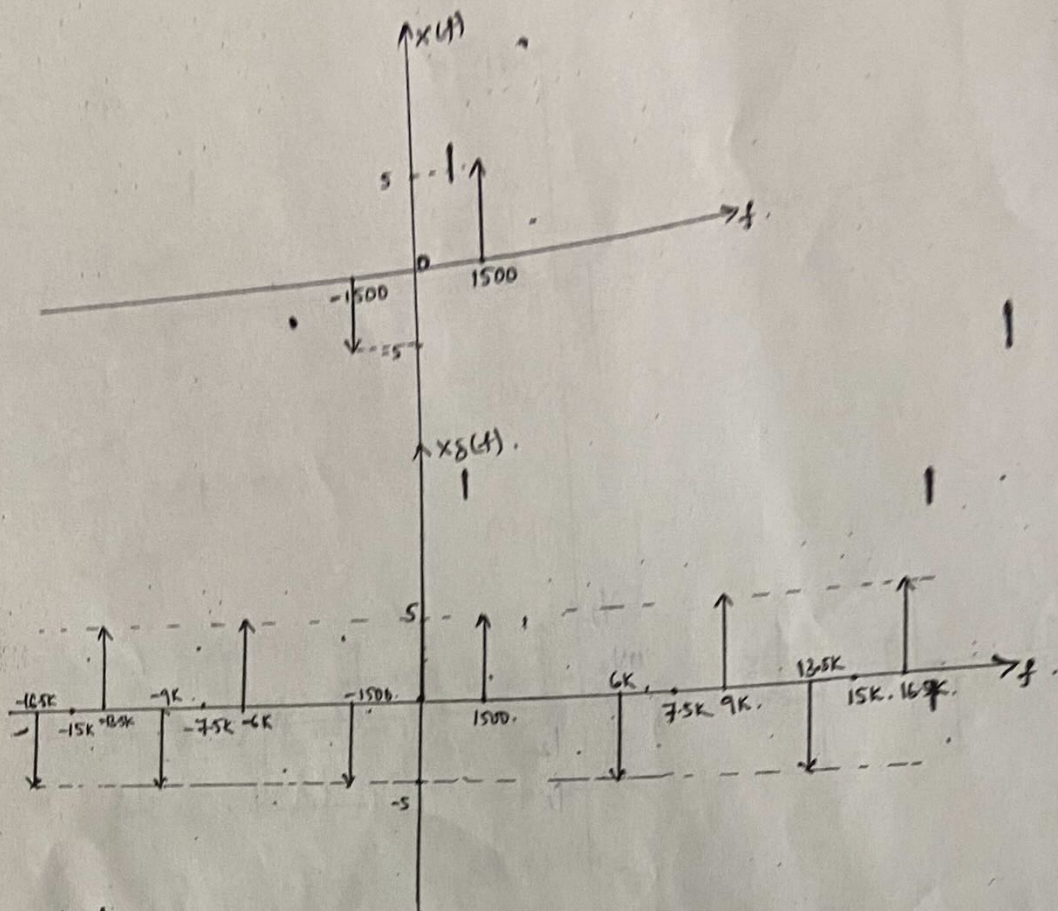
- ③. a waveform consists of sinusoidal with a second harmonic component of 1500Hz, with an amplitude of 5V. The signal is sampled at the rate of 7500 samples/sec ideally. Sketch the Spectrum of actual signal, also the spectrum of sampled signal showing first 3 components.

$$x(t) = 5 \sin(2\pi \times 1500t)$$

$$x(f) = \frac{5}{2j} [\delta(f-1500) - \delta(f+1500)]$$

sampled signal. $x_s(f) = f_s \sum_{n=-\infty}^{\infty} x(f-nf_s)$

$$x_s(f) = 7500 [x(f) + x(f \pm 7500) + x(f \pm 15000)]$$



- ④. Consider an Analog signal $g(t) = 3\cos 50\pi t + 10\sin 300\pi t - \cos 100\pi t$.
determine the Nyquist rate required for the signal & Nyquist interval

$$g(t) = 3\cos 50\pi t + 10\sin 300\pi t - \cos 100\pi t.$$

$$\text{Nyquist rate} = 300.$$

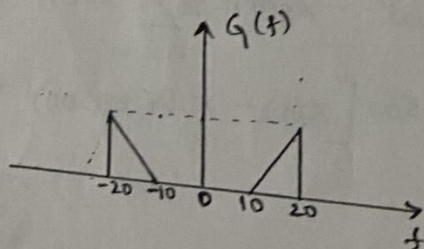
$$\text{Nyquist interval} = \frac{1}{300} = 3.33 \text{ msec}.$$

- ⑤. A Bandpass signal with spectrum shown in figure is ideally sampled. Sketch the Spectrum of Sampled signal when

① $f_s = 20 \text{ Hz}$.

② $f_s = 30 \text{ Hz}$.

③ $f_s = 40 \text{ Hz}$.



i)

520

(i) $f_s = 20 \text{ Hz}$, $f_{gu} = 20$, $f_{gl} = 10$.

$$f_s = 2 \frac{f_{gu}}{m} \Rightarrow m = \frac{2 f_{gu}}{f_s}$$

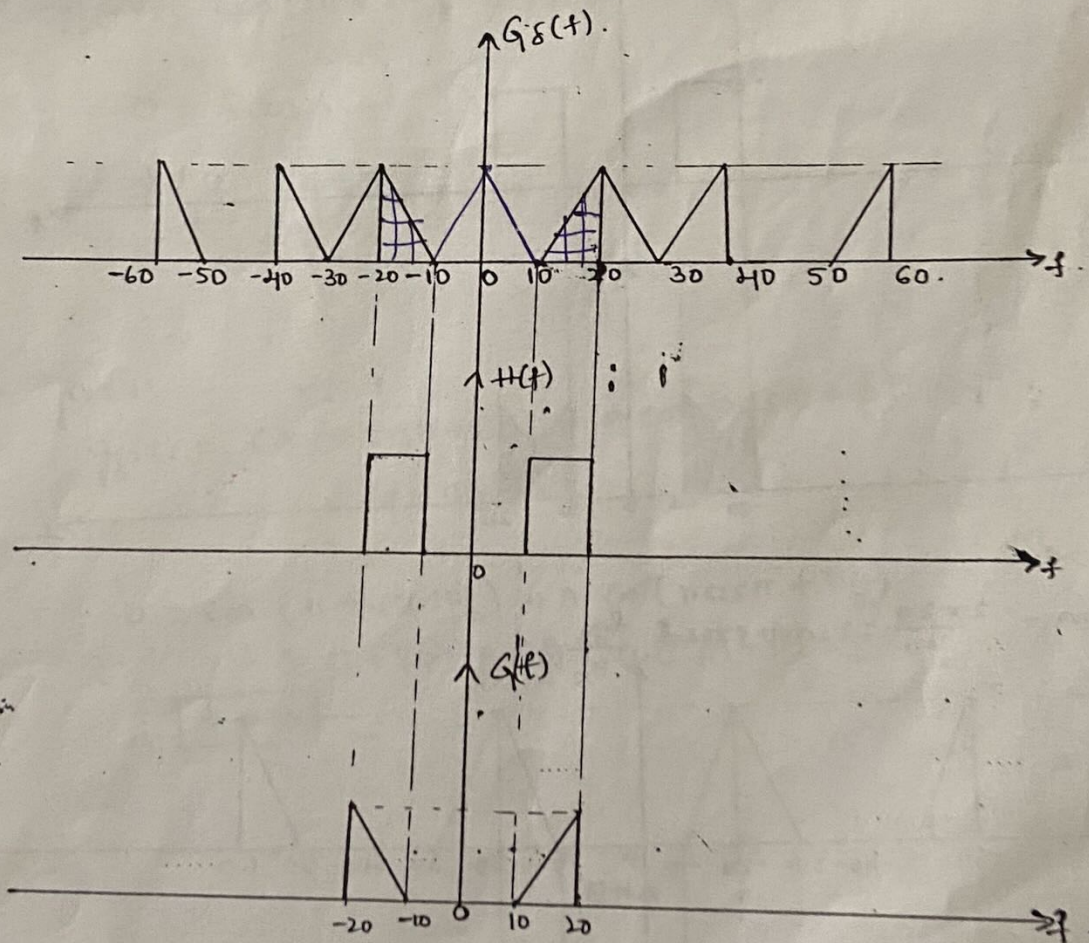
$$m = \frac{2 \times 20}{20}$$

$$m = 2,$$

$$B_x = f_{gu} - f_{gl} = 20 - 10 = 10$$

$$\frac{f_{gu}}{B_x} = \frac{20}{10} = 2$$

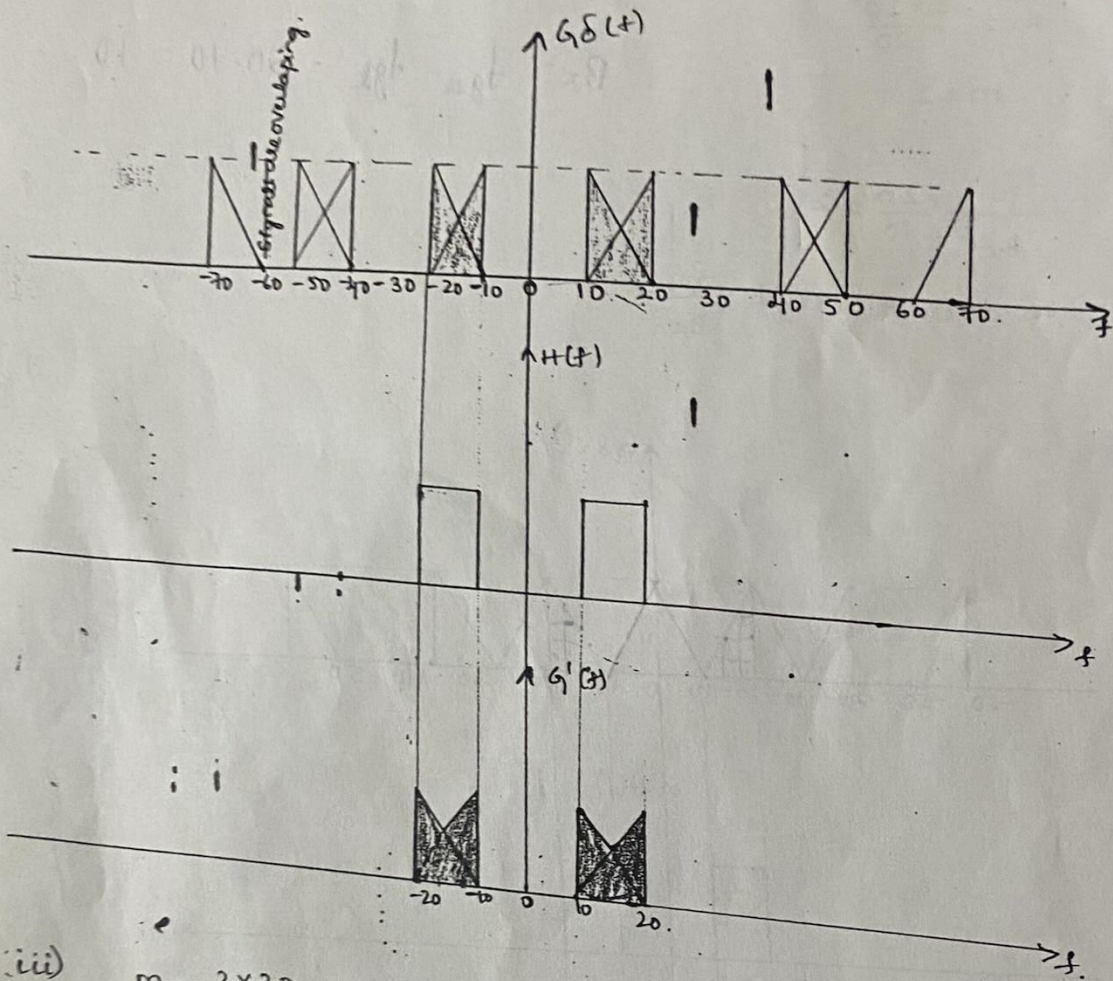
or does not exceed $\frac{f_{gu}}{B_x}$.



(ii). $m = \frac{2 \times 20}{30}$

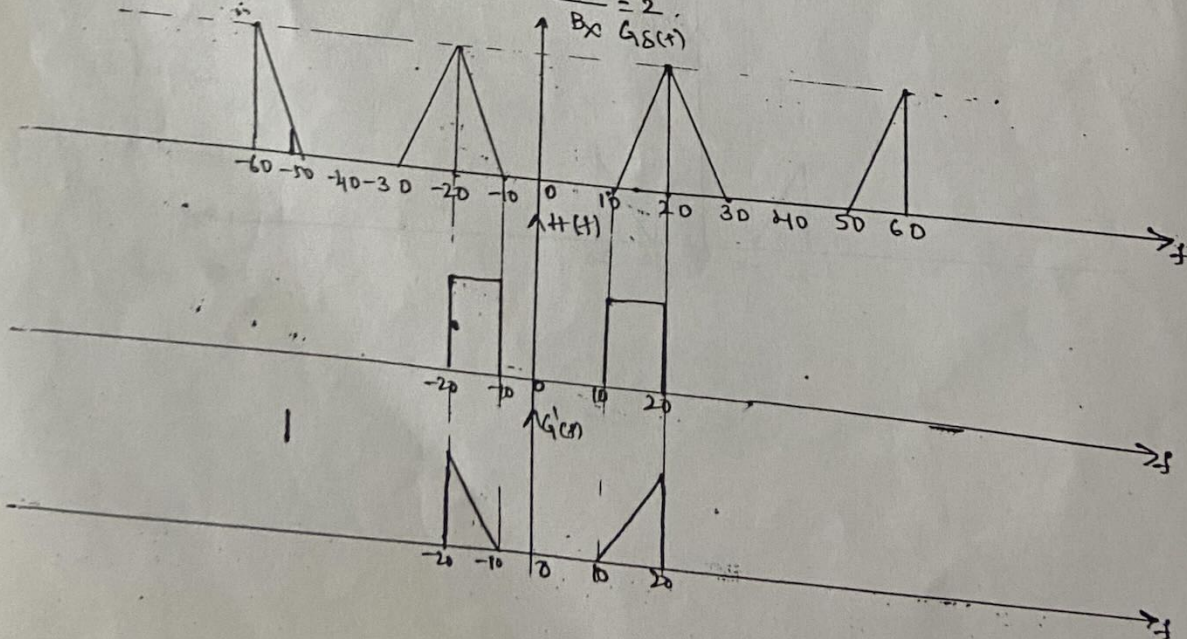
$m = 4/3$ not exceed $\frac{f_{gu}}{B_x} = 2$.

1.3



(iii)

$$m = \frac{2 \times 20}{40} = 1 \text{ not exceed } \frac{f_{gu}}{B_x G_{su}} = 2$$



- ⑥ a signal $g(t)$ consists of two frequency components $f_1 = 3.9 \text{ kHz}$ $f_2 = 4.1 \text{ kHz}$. in such a relationship that they just cancel each other when $g(t)$ is sampled at instants $t = 0, T_s, 2T_s, \dots$ where $T_s = 12.5 \mu\text{sec}$. The signal $g(t) = \cos(2\pi f_1 t + \pi/2) + A \cos(2\pi f_2 t + \phi)$ find the value of Amplitude & phase shift ϕ .

$$g_s(t) = \cos(2\pi \times 3.9 \text{ K} \times nT_s + \pi/2) + A \cos(2\pi \times 4.1 \text{ K} \times nT_s + \phi).$$

(i) at $n=0$.

$$g_s(t) = \cos(2\pi \times 3.9 \text{ K} \times 0 + \pi/2) + A \cos(2\pi \times 4.1 \text{ K} \times 0 + \phi)$$

$$0 = A \cos \phi.$$

$$A \neq 0, \quad \phi = \pi/2.$$

(ii). $n=1$.

$$g_s(t) = \cos(0.975\pi + \pi/2) + A \cos(1.025\pi + \phi)$$

$$\phi = \pi/2.$$

$$0 = \cos(0.975\pi + \pi/2) + A \cos(1.025\pi + \pi/2)$$

$$-0.078459 + A \times 0.078459 = 0$$

$$\boxed{A = 1V}$$

④. A signal $x(t) = 2 \cos 400\pi t + 6 \cos 640\pi t$ is ideally sampled at $f_s = 500\text{Hz}$. The sampled signal is passed through an ideal low pass filter with a cut-off frequency of 400Hz .

- Sketch the frequency spectrum of original signal.
- Sketch the frequency spectrum of sampled wave.
- State what frequency components will appear in O/P.

$$x(t) = 2 \cos 400\pi t + 6 \cos 640\pi t$$

$$= A_1 \cos 2\pi f_1 t + A_2 \cos 2\pi f_2 t$$

$$f_1 = 200$$

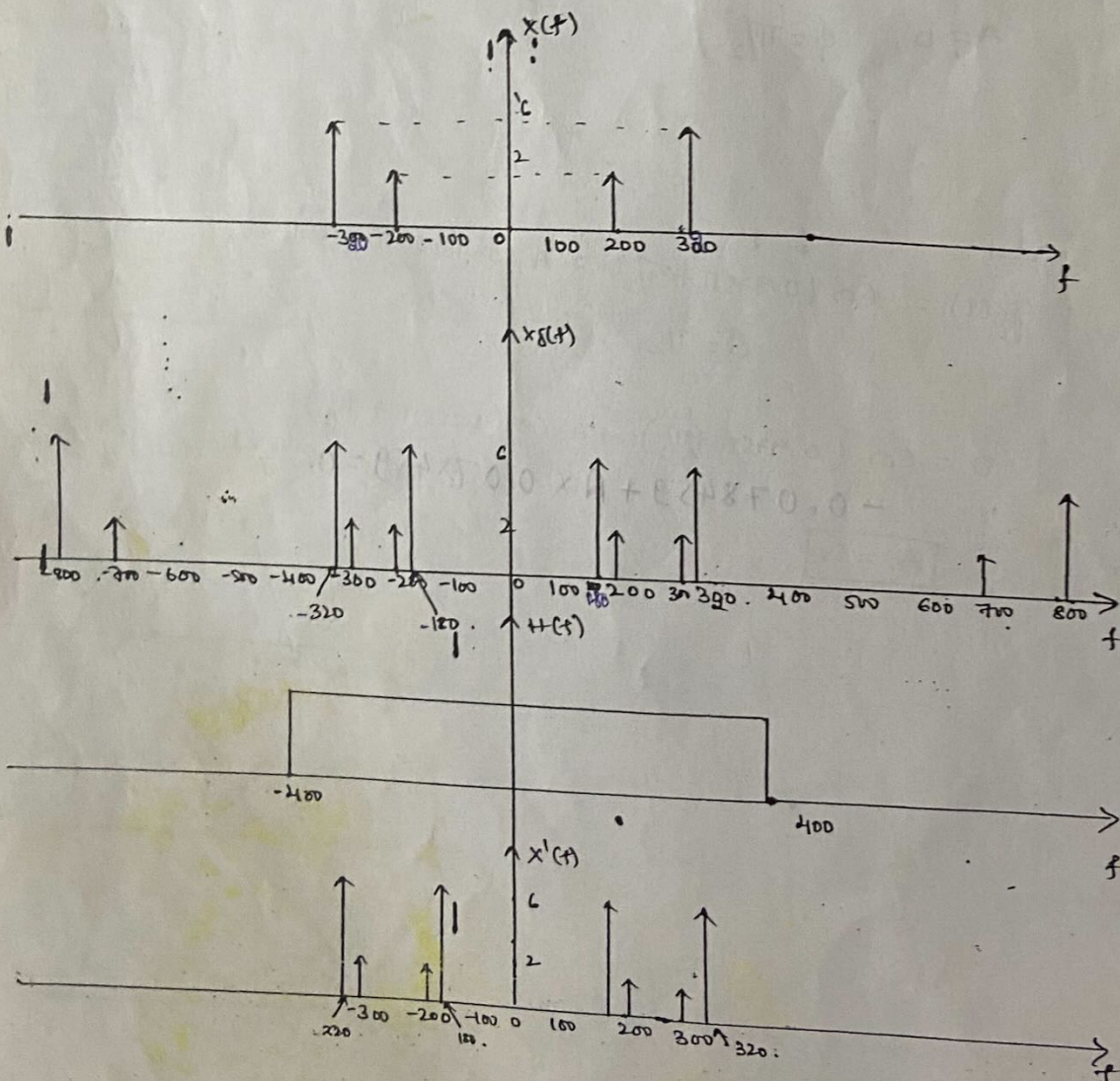
$$A_1 = 2$$

$$f_2 = 320$$

$$A_2 = 6$$

F.T. $\cos$

$$= \frac{1}{2} [\delta(f - f_c) + \delta(f + f_c)]$$



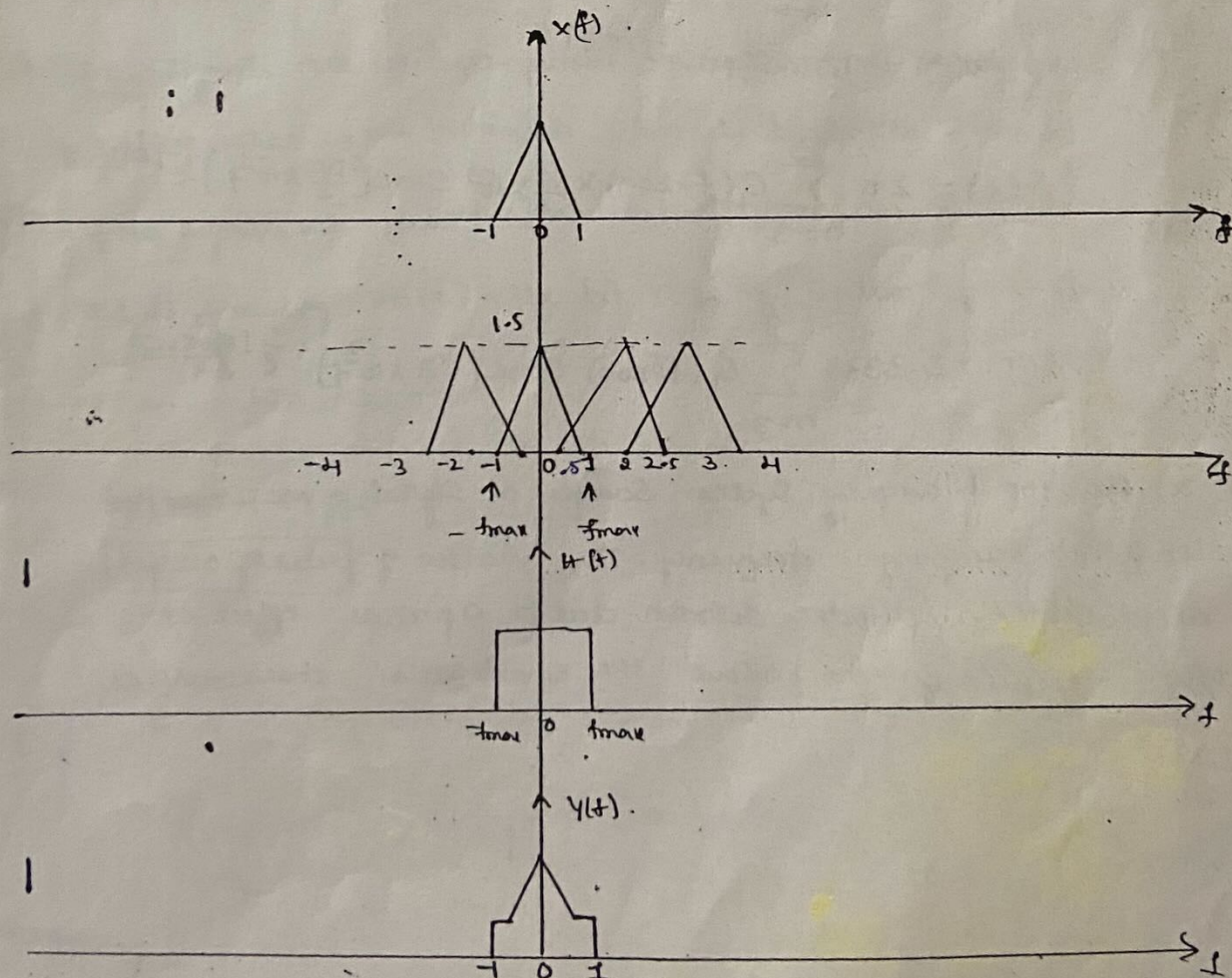
- ⑧. For the spectrum shown in figure the signal is sampled at the rate of $f_s = 1.5 f_{max}$. $f_{max} = 1\text{Hz}$. is maximum signal frequency. sketch the spectrum of sampled signal if the sampled signal is put through an ideal LPF of bandwidth f_{max} . Sketch the spectrum of output signal from this filter.

$$f_s = 1.5 f_{max}.$$

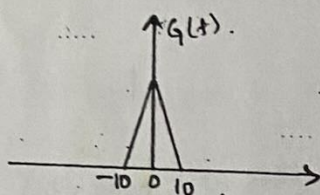
$$x_s(t) = f_s x(t) + \sum_{\substack{n=-\infty \\ n \neq 0}}^{\infty} f_s x(t - n f_s).$$

$$f_{max} = 1\text{Hz}.$$

$$x_s(t) = 1.5 x(t) + \sum_{\substack{n=-\infty \\ n \neq 0}}^{\infty} 1.5 x(t - 1.5 n)$$



- ⑨. The spectrum of signal $g(t)$ is shown in figure. The signal is sampled at the Nyquist rate with a periodic train of rectangular pulses of duration $\frac{50}{3}$ msec. find the spectrum of sampled signal for frequencies giving upto 30Hz with a relevant expression.



$$\omega = 10$$

$$f_s = 2\omega$$

$$= 2 \times 10$$

$$= 20 \text{ Hz}$$

$$\text{Nyquist rate} = \frac{50}{3} \times 10^{-3}$$

$$S(f) = f_s \sum_{n=-\infty}^{\infty} G(f - n f_s) \cdot H(f)$$

$$H(f) = T_s \text{sinc}(f T_s) \cdot e^{-j\pi f T_s}$$

$$S(f) = 20 \sum_{n=-3}^3 G(f - 20n) \times \frac{50}{3} \times 10^{-3} \text{sinc}\left(\frac{50}{3} \times 10^{-3} f\right) \cdot e^{-j\pi \frac{50}{3} \times 10^{-3} f}$$

$$S(f) = 0.3333 \sum_{n=-3}^3 G(f - 20n) \text{sinc}\left(\frac{50}{3} \times 10^{-3} f\right) \cdot e^{-j\pi \frac{50}{3} \times 10^{-3} f}$$

- ⑩. a flat-top / Sampling System samples a signal of maximum 1Hz with 2.5Hz sampling frequency. The duration of pulse is 0.2sec. calculate the amplitude distortion due to aperture effect at highest frequency. Also find out the equalization characteristics.

$$H(f) = T \operatorname{sinc}(fT) \cdot e^{-j\pi fT}$$

 $f_m,$

$$T = 0.2$$

$$H(f) = 0.2 \operatorname{sinc}(0.2f) \cdot e^{-j0.2\pi f}$$

$$|H(f)| = 0.2 \operatorname{sinc}(0.2f)$$

at $f = 1 \text{ Hz}$

$$|H(1)| = 0.18709$$

equalization characteristics

$$= \frac{K}{|H(f)|} \quad K=1$$

$$= 5.3400$$

$$\frac{0.2 \sin 180 \times 0.2}{0.2 \times \pi}$$

$$= 0.187$$

$$= 0.187$$

at $f = 2 \text{ Hz}$

- ⑪. A 24 voice signals are sampled uniformly and then time division multiplexed the highest frequency components for each voice signals is 3.4 kHz.

(i) if the signals are pulse amplitude modulated using Nyquist rate sampling what is the minimum channel bandwidth required

(ii). if the signals are pulse code modulated with an 8-bit encoder what is sampling rate? the bit rate of system is 1.5 Mbps.

$$(i) N = 24, \quad W = 3.4 \text{ kHz}$$

$$B_T = 24 \times 3.4 \text{ K}$$

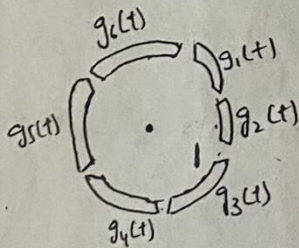
$$B_T = 81.6 \text{ K}$$

$$(ii) \text{ data rate of individual sample} = \frac{1.5 \text{ Mbps}}{24} = 62.5 \text{ K}$$

$$\text{Sampling rate} = \frac{62.5 \text{ K}}{8} = 7.8125 \text{ K}$$

- (12) 6 independent message sources of Bandwidth $w, w, 2w, 2w, 3w, 3w$ are to be transmitted on time division multiplexed basis using a common communication channel.
- (i) Setup a scheme for accomplishing this multiplexing requirement with each message signal is sampled at its Nyquist rate (ii) determine the minimum transmission Bandwidth of channel.

Signal	Bandwidth	Sampling rate.
$g_1(t)$	w	$2w$
$g_2(t)$	w	$2w$
$g_3(t)$	$2w$	$4w$
$g_4(t)$	$2w$	$4w$
$g_5(t)$	$3w$	$6w$
$g_6(t)$	$3w$	$6w$



(ii) $B_T = w + w + 2w + 2w + 3w + 3w$
 $= 12w$

Data rate $= 2 \times 12w$

Data rate $= 24w$

$B = \frac{1}{2} S, \quad S = 2 \times B$